Q1
What is the fundamental purpose of a computer network?

A
Sharing resources

B
Storing data

C
Performing calculations

D
Enhancing security

Q2
Which topology does the Internet resemble?

A
Ring

B
Bus

C
Star

D
Mesh

Q3
What type of network covers a large geographical area, like a city, country, or the world?

A
LAN

B
MAN

C
WAN

D
PAN

Q4
Which device is used to connect different network segments in a computer network?

A
Repeater

B
Router

C
Bridge

D
Modem

Q5
What protocol is commonly used for transmitting web pages over the internet?
A
SMTP
B
FTP
C
HTTP
D
SNMP

Q6
Which layer of the OSI model ensures the error-free transmission of data?
A
Physical Layer
B
Data Link Layer
C
Network Layer
D
Transport Layer

Q7
In a computer network, what is the main function of the application layer?
A
To provide network services to the applications
B
To transmit data between network devices
C
To package data for transfer
D
To route data between networks

Q8
Which command can be used to view the current IP configuration of a device?
A
ipconfig
B
ifconfig
C
Both ipconfig and ifconfig
D
netstat

Q9
What is the result of executing the ping localhost command?

A
It sends ICMP echo requests to the local machine

B
It configures the local machine's IP address

C
It displays the routing table

D
It clears the DNS cache

Q10
If a computer is unable to access the Internet, but can communicate with local network devices, what should be checked first?

A
Router configuration

B
Cable connections

C
Local firewall settings

D
DNS settings

Q11
A computer can ping IP addresses but not domain names.
What is likely the issue?

A
IP conflict

B
Router failure

C
DNS misconfiguration

D
Faulty Ethernet cable

Q12
Which layer of the OSI model is responsible for establishing, managing, and terminating sessions between applications?

A
Session Layer

B
Transport Layer

C
Application Layer

D
Presentation Layer

Q13
What is the primary function of the Network Layer in the OSI model?
A
To provide data routing paths for network communication
B
To format data
C
To establish connections
D
To encode and decode data

Q14
In the TCP/IP model, which layer corresponds to the OSI model's Physical and Data Link layers?
A
Application
B
Transport
C
Internet
D
Network Interface

Q15
The process of encapsulation involves data moving from which layer to which layer?
A
Application to Physical
B
Physical to Application
C
Transport to Network
D
Network to Transport

Q16
Which OSI layer is responsible for error detection and correction at the destination?
A
Transport
B
Network
C
Data Link
D
Physical

Q17
What does the TCP/IP model's Internet layer do?
A
Manages end-to-end data communication sessions
B
Routes packets across networks
C
Formats data packets
D
Encodes and decodes data

Q18
Which protocol operates at the Transport Layer of the OSI model to ensure reliable communication?
A
ICMP
B
IP
C
TCP
D
UDP

Q19
How does the Presentation Layer of the OSI model differ from the Application Layer?
A
It is responsible for session management
B
It deals with syntax and semantics of the information exchanged
C
It provides encryption and compression
D
It is not implemented in the TCP/IP model

Q20
What feature distinguishes TCP from UDP at the Transport Layer?
A
TCP is faster than UDP
B
TCP and UDP use different port numbers
C
TCP is connection-oriented while UDP is not
D
TCP uses IP but UDP does not

Q21
Which command can be used to display the current TCP/IP network configuration?
A
ipconfig /all
B
netstat -r
C
arp -a
D
tracert

Q22
What does the netstat command do?
A
Displays network connections, routing tables, and statistics
B
Configures network interfaces
C
Tests connectivity to a host
D
Manages firewall settings

Q23
Which command is used to test connectivity to a specific IP address and trace the path the packets take?
A
ping
B
arp
C
tracert
D
nslookup

Q24
A device is unable to connect to any network. Which OSI layer should be investigated first?
A
Application
B
Presentation
C
Network
D
Physical

Q25
If a computer can connect to local devices but not to the Internet, what might be misconfigured?

A
IP address
B
Subnet mask
C
Default gateway
D
MAC address

Q26
A web application can send data but cannot receive.
What layer could be malfunctioning?

A
Application
B
Session
C
Transport
D
Presentation

Q27
What is the primary function of the MAC (Media Access Control) sublayer?

A
Frame delimiting
B
Error detection
C
Controlling access to the physical medium
D
Physical addressing

Q28
What is the standard Ethernet frame size for most networks?

A
64 bytes to 1518 bytes
B
128 bytes to 1024 bytes
C
1500 bytes to 2000 bytes
D
100 bytes to 1500 bytes

Q29
In Ethernet networks, what is the purpose of the collision detection mechanism?
A
To encrypt data
B
To compress data
C
To detect and manage when two devices try to send data simultaneously
D
To prioritize data

Q30
Which Ethernet standard provides a speed of 1 Gbps?
A
10BaseT
B
100BaseTX
C
1000BaseT
D
10GBaseT

Q31
What type of cable is used for most modern Ethernet networks?
A
Coaxial cable
B
Fiber optic cable
C
Twisted pair cable
D
None of the above

Q32
What does the term "full-duplex" mean in the context of Ethernet networking?
A
Data can be transmitted in one direction only
B
Data can be transmitted in both directions, but not simultaneously
C
Data can be transmitted in both directions simultaneously
D
None of the above

Q33
How does a switch determine the destination of an Ethernet frame?
A
By using the source IP address
B
By using the destination MAC address
C
By using the source MAC address
D
By broadcasting to all ports except the source

Q34
What mechanism do Ethernet switches use to avoid collisions?
A
Collision detection
B
Collision prevention
C
Collision avoidance
D
Collision management

Q35
Which feature distinguishes managed switches from unmanaged switches in an Ethernet network?
A
The ability to encrypt data
B
The ability to be configured and monitored
C
The use of optical fibers
D
The support for multiple protocols

Q36
Which command can be used to display the MAC address of a network interface on a Windows machine?
A
ipconfig /all
B
ifconfig
C
arp -a
D
netstat -e

Q37
On a Linux system, which command shows the MAC address of all network interfaces?
A
ip addr
B
ifconfig -a
C
route
D
netstat -i

Q38
How can you change the MAC address of a network interface on Linux?
A
Modifying the network configuration file
B
Using the ifconfig command
C
Using the macchanger tool
D
All of the above

Q39
If a device is not receiving an IP address via DHCP over Ethernet, what layer should be investigated?
A
Application
B
Presentation
C
Network
D
Data Link

Q40
A computer can access local network devices but not the internet.
What should be checked on the switch?
A
VLAN configurations
B
Port speed settings
C
Power settings
D
Firmware updates

Q41
What could be a reason for slow data transfer rates over an Ethernet network?
A
Faulty NIC
B
Cable length exceeds standards
C
Incorrect duplex settings
D
All of the above

Q42
What is the primary purpose of the Internet Protocol (IP)?
A
To provide error checking and correction
B
To ensure reliable data delivery
C
To route packets across networks
D
To format data packets

Q43
What type of messages are used by ICMP to inform senders of network issues?
A
Query messages
B
Error-reporting messages
C
Routing-update messages
D
Service-quality messages

Q44
Which IP protocol version uses 128-bit addresses?
A
IPv4
B
IPv6
C
ARP
D
RARP

Q45
How do routers utilize ICMP messages within the IP framework?
A
To encrypt data packets
B
To prioritize packets
C
To diagnose routing issues
D
To compress data packets

Q46
What is the main difference between IPv4 and IPv6?
A
Address size
B
Error detection mechanisms
C
Encryption capabilities
D
Transport layer integration

Q47
Which feature of IPv6 eliminates the need for NAT (Network Address Translation)?
A
Auto-configuration
B
Built-in encryption
C
The vast address space
D
Multicast

Q48
What is the function of an IP subnet mask?
A
To separate the network address from the host address
B
To encrypt IP addresses
C
To detect IP conflicts
D
To assign IP addresses automatically

Q49
In IPv6, what is the equivalent of ARP (Address Resolution Protocol) used in IPv4?
A
DHCPv6
B
ICMPv6
C
Neighbor Discovery Protocol (NDP)
D
IGMPv6

Q50
What mechanism does IPv6 use to ensure data integrity and authentication?
A
ESP (Encapsulating Security Payload)
B
AH (Authentication Header)
C
ICMPv6
D
RARP

Q51
How can you display the current IP configuration on a Unix/Linux system?
A
ipconfig
B
ifconfig
C
netstat -i
D
arp -a

Q52
Which command can you use to test IPv6 connectivity to an address?
A
ping6
B
tracert6
C
arp -6
D
ifconfig -6

Q53
What command provides detailed information about routes to an IPv6 address and discovers MTU along the path?

A
mtr

B
tracepath6

C
ip -6 route

D
netstat -r6

Q54
A computer can access IPv4 websites but not IPv6 websites.
What should be checked first?

A
IPv6 configuration on the computer

B
Router's IPv6 support

C
IPv6 DNS settings

D
Firewall settings for IPv6

Q55
You're unable to ping an IPv6 address from your machine.
What could be the issue?

A
Incorrect subnet mask

B
Firewall blocking ICMPv6 packets

C
Incorrect gateway

D
All of the above

Q56
How to resolve intermittent connectivity issues with an IPv6 network?

A
Replacing the router

B
Updating network drivers and firmware

C
Checking for IPv6 prefix delegation issues

D
All of the above

Q57
What is the primary purpose of routing protocols in a network?
A
To encrypt data
B
To compress data
C
To determine optimal data paths
D
To manage user permissions

Q58
Which type of routing algorithm does OSPF use?
A
Distance-vector
B
Link-state
C
Path-vector
D
Hybrid

Q59
What differentiates dynamic routing from static routing?
A
Dynamic routing is less secure
B
Static routing automatically updates
C
Dynamic routing automatically updates routes
D
Static routing is faster

Q60
In BGP (Border Gateway Protocol), what is the main factor in selecting the best path?
A
Path length
B
Bandwidth
C
Lowest cost
D
Number of AS hops

Q61
How does a distance-vector routing protocol determine the best route?
A
By counting hops
B
By measuring latency
C
By bandwidth
D
By using a cost metric

Q62
What is the primary benefit of using a routing protocol like EIGRP compared to RIP?
A
Faster convergence
B
Higher security
C
Simpler configuration
D
Less bandwidth usage

Q63
Which routing protocol is classified as an Exterior Gateway Protocol (EGP)?
A
OSPF
B
EIGRP
C
BGP
D
RIP

Q64
What mechanism does OSPF use to prevent routing loops?
A
Split horizon
B
Route poisoning
C
Link-state advertisement
D
Area designations

Q65
In BGP, what is the purpose of the AS_PATH attribute?
A
To encrypt data packets
B
To specify the priority of routes
C
To record the path of ASes
D
To specify the route source

Q66
Which command can display the routing table on a Cisco router?
A
show ip route
B
display ip routing-table
C
route print
D
netstat -r

Q67
How can you manually add a static route to the routing table in Linux?
A
Using the route add command
B
Using the ifconfig command
C
Editing the /etc/hosts file
D
Using the ip route add command

Q68
In Cisco IOS, how can you redistribute OSPF routes into EIGRP?
A
By configuring redistribution in the OSPF process
B
By configuring redistribution in the EIGRP process
C
By using access lists
D
By using route maps

Q69
If a newly added network is not appearing in the routing tables, what should be checked first?
A
Physical connections
B
Route summarization settings
C
Access control lists
D
Network advertisements

Q70
A router is not choosing the expected path for traffic, what might be misconfigured?
A
Interface bandwidth settings
B
Routing protocol priorities
C
Next-hop IP addresses
D
All of the above

Q71
How can you resolve a persistent routing loop between two routers?
A
Increasing the hop count limit
B
Implementing hold-down timers
C
Correcting mismatched subnet masks
D
Configuring route filtering

Q72
What is the main difference between TCP and UDP?
A
TCP is connectionless, while UDP is connection-oriented
B
TCP is connection-oriented, while UDP is connectionless
C
TCP is faster than UDP
D
UDP is more secure than TCP

Q73
Which protocol is typically used for streaming media?
A
TCP
B
UDP
C
SCTP
D
RTP

Q74
How does TCP ensure data is delivered reliably and in the correct order?
A
By using error detection codes
B
By using sequence numbers and acknowledgments
C
By compressing data packets
D
By encrypting data packets

Q75
In TCP, what mechanism is used to control the rate of data transmission to prevent network congestion?
A
Error correction
B
Flow control
C
Data encryption
D
Data compression

Q76
What is the purpose of a TCP three-way handshake?
A
To establish a secure connection
B
To synchronize sequence numbers and acknowledge the connection
C
To compress data
D
To encrypt data

Q77
Which TCP header field is used for flow control?
A
Sequence Number
B
Acknowledgment Number
C
Window Size
D
Checksum

Q78
How does UDP handle error checking?
A
It uses sequence numbers and acknowledgments
B
It includes a checksum for error detection
C
It encrypts data packets
D
It compresses data packets

Q79
What is TCP's initial congestion window size typically set to?
A
1 MSS (Maximum Segment Size)
B
2 MSS
C
3 MSS
D
4 MSS

Q80
In TCP, what is the function of the congestion avoidance algorithm?
A
To reduce the data transmission rate before packet loss occurs
B
To encrypt data
C
To compress data
D
To ensure data is sent in order

Q81
How can you check open TCP ports on a Windows machine?

A
netstat -an

B
ipconfig /all

C
ifconfig -a

D
traceroute

Q82
In Linux, how can you view the UDP sockets currently in use?

A
netstat -au

B
ifconfig

C
ip addr

D
ss -u

Q83
How can you set a specific TCP window size when using the iperf command for testing?

A
iperf -w [size]

B
iperf -s

C
iperf -c [hostname]

D
iperf -l [length]

Q84
If a client is experiencing delayed TCP connections, what could be a potential cause?

A
Incorrect subnet mask

B
Slow DNS resolution

C
Incorrect routing

D
Firewall settings

Q85
A UDP-based application is experiencing packet loss.
What should be investigated first?
A
Application error handling
B
Network congestion
C
Incorrect port numbers
D
Firewall settings

Q86
How can you troubleshoot intermittent TCP connection timeouts?
A
Checking for faulty network hardware
B
Increasing the TCP timeout settings
C
Verifying network path stability
D
All of the above

Q87
What is the primary function of HTTP in the web architecture?
A
To encrypt data
B
To compress data
C
To transfer hypertext documents
D
To stream video content

Q88
FTP uses which mode for establishing data connections by default?
A
Active mode
B
Passive mode
C
Secure mode
D
Anonymous mode

Q89
Which of the following HTTP methods is idempotent and used for updating resources on a server?

A
GET
B
POST
C
PUT
D
DELETE

Q90
In HTTP, what status code indicates that the requested resource has been permanently moved to a new URL?

A
200 OK
B
301 Moved Permanently
C
404 Not Found
D
500 Internal Server Error

Q91
What is the main advantage of using FTP over HTTP for file transfer?

A
Higher security
B
Better compression
C
Support for directory listing and manipulation
D
Faster data transfer

Q92
What mechanism does HTTP/2 use to reduce latency and improve load times compared to HTTP/1.1?

A
Data compression
B
Multiplexing
C
Encryption
D
Tokenization

Q93
How does FTP ensure data is transferred securely?
A
By using SSL/TLS for encryption
B
By compressing data
C
By using checksums for integrity
D
By using UDP

Q94
Which feature of HTTP/3 aims to improve performance over lossy or unreliable networks?
A
QUIC protocol
B
WebSocket support
C
Server Push
D
Content-Encoding

Q95
In FTP, what is the difference between binary and ASCII transfer modes?
A
Binary mode transfers files as text, while ASCII mode transfers them as binary
B
Binary mode compresses data, while ASCII mode does not
C
Binary mode transfers files in a binary format without conversion, while ASCII mode converts files between different text formats
D
Binary mode is more secure than ASCII mode

Q96
How can you download a file from an FTP server using the command line in Linux?
A
wget [FTP URL]
B
curl -O [FTP URL]
C
ftp [FTP URL]
D
scp [FTP URL]

Q97
How can you send a custom HTTP header with a curl request?
A
curl -d "[Header]" [URL]
B
curl -H "[Header]: [Value]" [URL]
C
curl --header "[Header]" [URL]
D
curl --custom-header "[Header]: [Value]" [URL]

Q98
What command allows you to interactively communicate with an HTTP server using telnet?
A
telnet [host] 80 followed by GET / HTTP/1.1
B
telnet [host] 443 followed by CONNECT / HTTP/1.1
C
telnet [host] 80 followed by POST / HTTP/1.1
D
telnet [host] 80 followed by HEAD / HTTP/1.1

Q99
If a web page is not loading correctly, what should be checked first?
A
Browser cache
B
Server configuration
C
DNS settings
D
Network connectivity

Q100
A file transfer via FTP fails with a permission error.
What is likely the cause?
A
Incorrect username or password
B
Network congestion
C
File not found on the server
D
Firewall settings

Q101
How can you diagnose slow FTP file transfers?
A
Checking file size and type
B
Analyzing network bandwidth and congestion
C
Verifying FTP server load
D
All of the above

Q102
What is the primary purpose of a firewall in network security?
A
To detect and remove viruses
B
To manage network traffic based on rules
C
To encrypt data transmissions
D
To authenticate users

Q103
Which cryptographic algorithm is commonly used for secure password storage?
A
RSA
B
AES
C
SHA-256
D
DES

Q104
What is the function of a digital certificate in network security?
A
To ensure data integrity
B
To verify the identity of entities exchanging information online
C
To compress data
D
To increase data transfer speed

Q105
What differentiates symmetric from asymmetric encryption?
A
The use of one key for encryption and a different key for decryption
B
The speed of encryption and decryption
C
The type of algorithms used
D
The use of a single key for both encryption and decryption

Q106
In the context of cryptography, what is a nonce?
A
A unique number used once
B
A method for encrypting data
C
A type of digital signature
D
A protocol for secure communication

Q107
How does a public key infrastructure (PKI) enhance network security?
A
By providing digital certificates that verify the ownership of public keys
B
By encrypting all data on the network
C
By detecting network intrusions
D
By managing network traffic

Q108
What is the primary security concern addressed by HTTPS?
A
Data corruption
B
Identity theft
C
Data interception during transmission
D
Unauthorized access to network resources

Q109
What role does the Diffie-Hellman algorithm play in cryptography?
A
It provides a method for digital signatures
B
It is used for secure password storage
C
It enables the secure exchange of encryption keys over a public channel
D
It encrypts and decrypts data

Q110
How can you generate a 2048-bit RSA key pair using OpenSSL?
A
openssl genpkey -algorithm RSA -out private_key.pem 2048
B
openssl rsa -genkey -out private_key.pem 2048
C
openssl genrsa -out private_key.pem 2048
D
openssl rsakey -create -size 2048 -out private_key.pem

Q111
What command can be used to encrypt a file using AES-256-CBC with OpenSSL?
A
openssl aes-256-cbc -in plaintext.txt -out encrypted.txt -pass pass:your_password
B
openssl encrypt -aes-256-cbc -in plaintext.txt -out encrypted.txt -k your_password
C
openssl enc -aes-256-cbc -in plaintext.txt -out encrypted.txt -k your_password
D
openssl aes-256-cbc -e -in plaintext.txt -out encrypted.txt -k your_password

Q112
If an encrypted email cannot be decrypted by the recipient, what should be checked first?
A
The email client's compatibility with the encryption standard
B
The correctness of the recipient's public key
C
The strength of the encryption algorithm
D
The sender's digital signature

Q113
How can network administrators detect and mitigate a man-in-the-middle (MitM) attack?
A
Implementing stronger encryption algorithms
B
Regularly updating firewall rules
C
Using digital certificates and HTTPS
D
Using network monitoring tools

Q114
Which frequency band is not commonly used by Wi-Fi networks?
A
2.4 GHz
B
5 GHz
C
900 MHz
D
60 GHz

Q115
What is the primary purpose of the Mobile IP protocol?
A
To secure wireless communications
B
To manage mobile device power consumption
C
To enable devices to move across networks without losing connectivity
D
To compress data for faster transmission

Q116
In the context of mobile computing, what is edge computing?
A
Processing data in a central data center
B
Storing data exclusively on the mobile device
C
Processing data near the source of data generation
D
Using cloud storage as the only data storage option

Q117
Which technology underpins the LTE (Long Term Evolution) standard for mobile communication?

A
GSM

B
CDMA

C
OFDMA

D
TDMA

Q118
How do MIMO (Multiple Input Multiple Output) technologies improve wireless network performance?

A
By encrypting data transmissions

B
By increasing the range of frequencies used

C
By using multiple antennas for transmission and reception

D
By compressing the data transmitted

Q119
What challenge does the hidden node problem pose in wireless networks?

A
Difficulty in achieving synchronization

B
Increased risk of data interception

C
Interference and collision of data transmissions

D
Excessive energy consumption

Q120
What is the main advantage of using SDN (Software Defined Networking) in wireless networks?

A
Enhanced security against cyber threats

B
Simplified hardware requirements

C
Flexible network management and configuration

D
Increased data transfer speeds